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Quantization-Aware Digital PID Control of an Inverting Buck-Boost Converter: A Simulation-to-Hardware Feasibility Audit on ESP32-S3 and STM32F103

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ABSTRACT Digital control of switching power converters is shaped by three implementation-quantized quantities: the ADC resolution of the sensed output, the DPWM resolution of the commanded duty, and the sampling period at which controller and power stage are synchronized. Whether the resulting design limits are properties of the hardware or of the software stack is rarely asked. This feasibility audit asks that question for an inverting buck-boost converter (Vref $= -12\,\mathrm{V}$, $\mathrm{f_{sw}} = 100\,\mathrm{kHz}$). The paper plant is ideal ($\mathrm{L} = 100\mu\mathrm{H}$, $\mathrm{C} = 220\mu\mathrm{F}$, $\mathrm{R} = 10\Omega$, $\mathrm{R_L} = \mathrm{ESR} = 0$, $\mathrm{f_0} \approx 1.07\,\mathrm{kHz}$); the as-wired hardware plant is $5\Omega + 330\mu\mathrm{H}{+}2.2\mu\mathrm{F}$ ($\mathrm{f_0} \approx 5.9\,\mathrm{kHz}$, $\mathrm{Q} \approx 2.45$) plus LM358 follower. A 79-run switched-model study on the paper plant with zero delay first locates the critical sampling boundary near 3.5–$4{\times}\mathrm{T_{sw}}$ (35–$40\,\mu\mathrm{s}$) and shows the three quantizers interact super-additively; delay- ($65\mu\mathrm{s}$ ESP Arduino, $11\mu\mathrm{s}$ STM32 bare) and noise- ($\sigma \approx 1.65/\sqrt{\mathrm{N}}\,\mathrm{V}$) augmented re-runs explain why the $2.2\mu\mathrm{F}$ hardware never settles with frozen gains even at $1\times$ (Fig. 20). The same controller is ported, from one shared firmware source, to ESP32-S3 (240 MHz, FPU) and STM32F103 (72 MHz, no FPU). The central finding is an abstraction-layer tax: through Arduino both MCUs are ADC-bound (ESP32-S3 $60\mu\mathrm{s}$, $73\mu\mathrm{s}$ floor $7.3 \times \mathrm{T_{sw}}$; STM32 $72.8\mu\mathrm{s}/23.9\mu\mathrm{s}$, $232\mu\mathrm{s}$ soft-float quantizer), but STM32 bare recovers $11.0\mu\mathrm{s}$ ADC (first-sample $97\mu\mathrm{s}$, $17\times$ ideal) and $39.8\mu\mathrm{s}$ paced floor ($4.0\times$) vs ESP $70.8\mu\mathrm{s}$ — still inside the paper-plant stable region while ESP is deep outside. Baseline hardware is pacing/noise metrology on the jumper surrogate; closed-loop as-wired validation (§V-G) shows Mode T (paced, single raw, frozen gains) never settles on the LC (447ms window-clip) while Mode S (averaged $\mathrm{N} = 1{+}\mathrm{EMA}$) settles at 3.6ms, and an adaptive $\mathrm{N} = 8 \leftrightarrow 64$ demo on the surrogate cuts updates 84% (0.98 on the LC) while holding $\pm 40\mathrm{mV}$. All data, firmware and $220\mu\mathrm{F}$ vs $330\mu\mathrm{H}$ simulations are in the companion repo.

INDEX TERMS digital PID control, quantization, buck-boost converter, ESP32-S3, STM32F103, cross-platform, adaptive sampling, measurement latency

I. INTRODUCTION

Digital control loops are now standard in switching power converters, driven by programmability, immunity to analog component drift, and integration with digital loads [1], [2]. The complete digital loop contains two unavoidable quantizers: the analog-to-digital converter (ADC) that samples the output voltage, and the digital pulse-width modulator (DPWM) that resolves the duty ratio. Their resolutions, together with the sampling period of the controller, jointly determine regulation quality. Classic work established that coarse quantization causes steady-state limit cycling and conditions between DPWM and ADC resolution for its elimination [3], [4], [5], while quantization-effect modeling permits quantitative prediction of these oscillations [6]. Recent work extends the analysis to MHz-range GaN converters using dither and sigma-delta modulation [7], [8], [9] and to design guidelines for two-loop current-mode digital controllers [8].

Digital control relieves the analog constraints of passive compensation, enables programmability and auto-tuning, and integrates naturally with digital loads [1], [2]. In exchange, the designer accepts three implementation-quantized quantities that an analog loop never sees: the ADC resolution of the measured output, the DPWM resolution of the commanded duty, and the sampling period at which controller and power stage are synchronized; DPWM-architecture and delay considerations are central to the achievable resolution [10], [11].

The theoretical basis is well established. Peterchev and Sanders derived conditions on ADC/DPWM resolution trade-offs for limit-cycle elimination and introduced digital dither to recover effective DPWM resolution [3]. Peng, Prodić, Alarcón, and Maksimović derived static and dynamic quantization models explaining the origins of limit-cycle oscillations [4], and Maksimović and Zane provided the small-signal discrete-time framework used to design digitally controlled converters [6]. A fuller survey appears in [12]. On the application side, low-resolution non-uniform ADC quantization has been shown to improve dynamic response [13], and the full control loop benefits from predictive structures [14], [15].

Two observations motivate this work. First, most quantization studies isolate a single stage, whereas a practical controller is constrained in all three quantities simultaneously; the interaction between sampling period, ADC word length, and PWM word length is rarely treated as one design problem. Second, the buck-boost converter, whose inverted output and non-minimum phase control path [16] make regulation inherently harder, is underrepresented in the digital-control literature: most buck-boost-specific digital studies consider time-domain or PID structures at a single resolution [17], [18] and do not sweep the full implementation space. The classical converter model itself is textbook material [19].

The central question of this work is therefore: how do sampling period, ADC quantization, PWM resolution, and real-time computational latency jointly affect the closed-loop performance of a microcontroller-controlled inverting buck-boost converter, and can a lightweight quantization-aware adaptive PID maintain acceptable regulation while reducing unnecessary controller-update effort — and do the answers depend on the microcontroller? Answering it requires three levels of validation that earlier studies treat separately: a continuous (analog) baseline, a digital simulation that introduces sampling, ADC and PWM quantization and computational delay, and surrogate (jumper-emulated) measurements on two embedded platforms. The microcontrollers are not merely devices that run the PID; they are the experimental vehicles for exposing quantities a simulator cannot produce directly: the acquisition time of the on-chip ADC, the execution time of the control step (with and without a hardware FPU), the update latency of the PWM peripheral, and the resulting control-loop latency and CPU utilization. The study uses one shared firmware source on an ESP32-S3 (240 MHz, Arduino core, LEDC + `analogRead`) and an STM32F103 Blue Pill (72 MHz, no FPU, STM32duino), whose 17× bare-register ADC advantage isolates whether the predicted sampling boundary is a hardware fact or a firmware artifact.

This paper reports a simulation-to-hardware feasibility audit of a digitally controlled inverting buck-boost converter under simultaneous quantization of sampling period, ADC resolution, and PWM resolution. The scope is explicitly that of an audit, not a completed converter regulation claim: the simulated plant is ideal and zero-delay, and baseline hardware is pacing/noise metrology on a jumper surrogate while the as-wired LC 330 μH plant (§V-G) provides the first closed-loop LC data. Contributions: (1) a 79-run factorial study on the paper plant that separates quantizer effects, locates the 3.5–4× (35–40μs) boundary and quantifies super-additivity (window-clip limited); (2) objective worst-case selection post-sweep; (3) half-scale coincidence $V_{ref} = -FS/2$ masking $|e_{ss}|$ at low ADC; (4) ESP32-S3 latency decomposition (ADC 60.0μs, PID 0.86μs int32, PWM 4.2μs, $T_{lat} \approx 65\mu$s, paced 70.8–73μs 7.1–7.3×); (5) STM32F103 cross-platform replication (Arduino 72.8/23.9μs, bare 11.0μs avg/4.23μs ideal, 5.88μs PWM, 232μs soft-float quantizer, 6.6× field / 17× ideal ADC tax, Table 6); (6) pacing sweep (mode T, 1–5×) validating 26μs (2.6×, jumper) and 39.8μs (4.0×, LC 330 μH) floors on STM32 bare vs 70.8–73μs on ESP, without claiming LC regulation in T; (7) effective-resolution experiment showing 1.6–1.9V noise floor masks quantizers; (8) full N = 1/8/32/64 ladder on the as-wired LC 330 μH (Table 7: 3.6–75ms, vs T 447ms) and STM32 LC (Table 8) closing the N = 1-only gap; (9) delay/noise-augmented re-sims reconciling sim with hardware. Novelty is the integrated stack-tax audit from one firmware, not sampling/dithering alone.

II. SYSTEM MODEL AND DIGITAL CONTROL

The paper plant is an ideal inverting buck-boost converter (Fig. 1) with input $V_{in} = +12$ V, reference $V_{ref} = -12$ V, $L = 100\,\mu$H, $C = 220\,\mu$F, switching frequency $f_{sw} = 100$ kHz ($T_{sw} = 10\,\mu$s), and load $R = 10\,\Omega$ ($f_0 \approx 1.07$ kHz). The as-wired hardware plant validated in §V-G is $5\Omega + 330\,\mu$H$+2.2\,\mu$F ($f_0 \approx 5.9$ kHz, $Q \approx 2.45$) plus LM358 follower (Fig. 8; design showed $100\,\mu$H, $f_0 \approx 10.7$ kHz, $Q \approx 1.3$). The 2.2μF plant is 5–10× faster than the paper plant and, as §VI and Fig. 20 show, its 35–40μs boundary does not transfer without retuning.

During OFF, the inductor current is held at zero when it would otherwise reverse (zero-current boundary), which counts discontinuous-mode (DCM) cycles and keeps the model valid in DCM without clamping the output. Many simplified studies go silent on this point. All reported results are obtained on this switched-

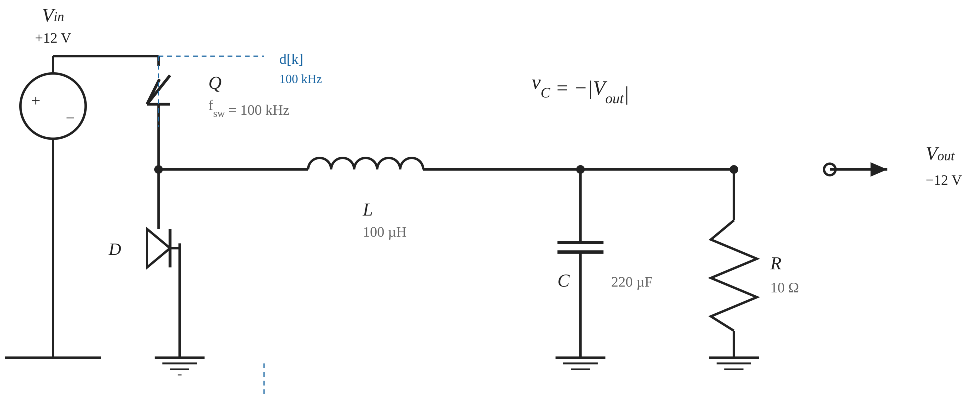

FIGURE 1. **Inverting buck-boost power stage used in the study (V_{in} = +12 V, L = 100 μH, C = 220 μF, f_{sw} = 100 kHz, R = 10 Ω; output $V_{out} \approx -12$ V at $D = 0.5$).**

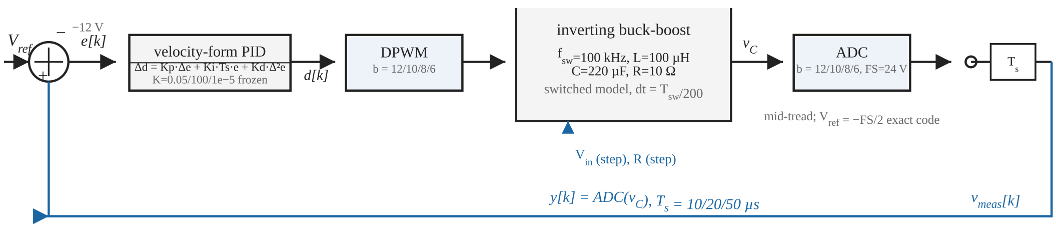

FIGURE 2. **Digital control loop: velocity-form PID, DPWM and ADC quantizers ($b \in \{12, 10, 8, 6\}$ bit, FS = 24 V), and sampling at $T_s = 10/20/50\ \mu$s. The reference sits exactly at the half-scale code $V_{ref} = -FS/2$.**

state model, not on an averaged approximation, so quantization artifacts (limit cycles, DCM chatter) appear faithfully; nonideal DCM of the buck-boost requires explicit boundary modeling [20].

$$\mathrm{ON:}\quad \frac{di_L}{dt} = \frac{+V_{in}}{L}, \quad \frac{dv_C}{dt} = -\frac{v_C}{RC} \tag{1}$$

$$\mathrm{OFF:}\quad \frac{di_L}{dt} = \frac{v_C}{L}, \quad \frac{dv_C}{dt} = -\frac{i_L + v_C/R}{C} \tag{2}$$

A. DIGITAL CONTROLLER AND QUANTIZATION

A velocity-form digital PID with positive gains (Fig. 2) operates at the sampling period T_s:

$$\begin{aligned} d[k] &= d[k-1] + K_p\big(e[k] - e[k-1]\big) + K_i T_s\, e[k] \\ &\quad + \frac{K_d}{T_s}\big(e[k] - 2e[k-1] + e[k-2]\big) \end{aligned} \tag{3}$$

with $e[k] = v_{meas}[k] - V_{ref}$, duty saturated to [0.05, 0.95], and no separate integrator state (inherent anti-windup). The measurement, duty, and update cadence are each quantized independently:

- Sampling: the controller updates every $T_s \in \{T_{sw}, 2T_{sw}, 5T_{sw}\}$ (10, 20, 50 μs);
  ADC: mid-tread quantizer, full-scale FS = 24 V, 12/10/8/6 bits. Note $V_{ref} = -FS/2$, so the reference is exactly representable at every resolution;
  PWM: duty quantized to 2^b levels, $b \in \{12, 10, 8, 6\}$.

A continuous-ideal baseline recomputes the same PID every dt with no quantization (true analog reference). The frozen controller gains were obtained once, deterministically, before any quantization study: grid search over $K_p \in \{0.005, 0.01, 0.02, 0.05\}$, $K_i \in \{20, 100, 500, 2000\}$, $K_d \in \{0, 2\times10^{-6}, 10^{-5}\}$ with cost $J = o_v/10 + t_s/5\,\mathrm{ms} + \mathrm{IAE}/0.05$ under constraints overshoot $\le 10\%$ and $|e_{ss}| \le 60\,\mathrm{mV}$, re-validated on the switched model: $K_p = 0.05$, $K_i = 100.0$, $K_d = 10^{-5}$. No parameter was touched during the quantization study.

III. SIMULATION FRAMEWORK AND METHODOLOGY

The core 79 simulations are on the paper plant (100μH, 220μF, 10Ω, $f_0 \approx 1.07$ kHz). They comprise: the continuous baseline, the ideal digital controller ($T_s = T_{sw}$, 16-bit ADC, 16-bit PWM), three sampling-only points, a six-point focused sampling sweep across the critical boundary, four ADC-only, four PWM-only runs, the full $3\times4\times4$ factorial sweep (48 runs), six disturbance runs (120 ms total, step at $t = 60$ ms, evaluated over the 60 ms post-step window, 6000 T_{sw}, on the continuous baseline, the ideal digital controller, and the objectively selected worst configuration), and six operating-point robustness runs ($V_{in} = 10, 15$ V on three configurations). Audit extensions re-run the same controller on the as-wired plant (330μH, 2.2μF, 5Ω, $f_0 \approx 5.9$ kHz) and with injected pipeline delay (65μs ESP Arduino, 11μs STM32 bare) and acquisition noise ($\sigma \approx 1.65/\sqrt{N}$ V) via `sim_core.BuckBoost(comp_delay_s, adc_noise_sigma)`; see §VI, §VI-A and §V-G. Solver: fixed-step Euler dt $= 1/(f_{sw} \cdot 200) = 0.05\mu$s (200 steps/$T_{sw}$), ZOH at period start, zero delay in the 79-run core, $r_{DCM} = N_{DCM}/N_{sw} \leq 1$ asserted.

Metrics: 2% settling time, overshoot, steady-state error, peak-to-peak ripple (limit cycle), IAE, ITAE, and N_{DCM} (switching periods containing a DCM event, $r_{DCM} = N_{DCM}/N_{sw} \leq 1$). For disturbance runs all metrics are evaluated on the 60 ms post-step window (6000 T_{sw} within a 120 ms total run) and ripple on its final 50%. The worst configuration is selected objectively after the sweep, as the run with the largest normalized degradation of steady-state error, IAE, and limit cycle relative to the ideal digital controller. No configuration is pre-declared worst.

IV. RESULTS

A. CONTINUOUS VERSUS IDEAL DIGITAL

The conversion from analog to ideal digital control already costs measurable quality (Table 2): the 16-bit/one-cycle system settles 36% slower (2.66 vs 1.96 ms) and shows a 12.76 mV steady-state offset and 39.6 mV peak-to-peak ripple against the continuous 0.44 mV / 26.7 mV. The 12.8 mV residual offset is sampling/ZOH and numerical error for these gains (sampling dominates at 16-bit, not quantization alone); every further result is expressed relative to it.

B. INDIVIDUAL STAGE EFFECTS

Fig. 3 isolates the sampling stage at 16-bit resolution and locates the critical boundary. Settling degrades smoothly as T_s/T_{sw} rises from 1 to 3.5 (2.66 to 6.38 ms, all still regulation-valid), then the loop loses regulation outright at 4×: it never settles, the peak-to-peak ripple jumps to 1.59 V, and 623 switching periods containing a DCM event accrue in the 60 ms window (6000 T_{sw}, $r_{DCM} = 0.10$). The practical boundary therefore sits between 3.5× and $4 \times T_{sw}$ (35–40 μs); $T_s = 5T_{sw}$ (50 μs) fails in every configuration, alone or combined.

ADC resolution couples to the reference in a subtle way (Fig. 4): because $V_{ref} = -FS/2$ is exactly representable, the 8-bit and 6-bit ADCs lock onto the exact code and reduce the measured steady-state error to ≈0.44 mV while masking a worse transient (overshoot rises from 0.06% at 12-bit to 0.85% at 6-bit; settling from 2.70 to 3.48 ms). This is a half-scale coincidence, not an ADC benefit.

PWM resolution below 8 bits dominates the limit cycle (Fig. 5): 6-bit PWM adds a 204.8 mV ripple versus ≈39.5 mV at 12/10/8-bit. Duty quantization is the direct source of steady-state ripple, as predicted by limit-cycle analysis [3], [4]; at 6-bit the duty grid step is 1 LSB $= 1/64$ over $[0, 1]$ (clamped to $[0.05, 0.95]$; over the clamped 0.9 range this would be $0.9/63 \approx 0.0143$ and 0.69 V static, 9% lower), and the loop settles into a limit cycle toggling between the adjacent codes 31/64 and 32/64 around the $D \approx 0.5$ operating point, whose measured peak-to-peak output excursion is 204.8 mV. The static code-to-code shift at this operating point is larger, $|dV_{out}/dD| \cdot 1/64 = 0.75$ V with $|dV_{out}/dD| = V_{in}/(1-D)^2 = 48$ V per unit duty at $D = 0.5$; the dynamic ripple is smaller because the output capacitor averages the excursion over the limit-cycle period. This is precisely the regime addressed by finer DPWM controller implementations [11], [10] and by noise-shaping modulators [21].

C. COMBINED SWEEP AND INTERACTIONS

In the 48-run factorial sweep the worst configuration is $T_s = 5T_{sw}$, ADC=8-bit, PWM=12-bit (normalized degradation score 1309.2): settling never completes within 60 ms, steady-state error −13.43 V, peak-to-peak ripple 7.33 V, IAE 0.789 V·s, 884 switching periods containing a DCM event ($r_{DCM} = 0.15$, 6000 T_{sw}). Fig. 6 classifies every combination against explicit thresholds (settling ≤ 6 ms, $|e_{ss}| \leq 120$ mV, limit cycle ≤ 120 mV, overshoot, DCM) into excellent / acceptable / marginal / unusable. Sampled at $T_s = T_{sw}$, 13 of 16 combinations are excellent and the rest acceptable; at 2× all are acceptable or marginal; at 5× all 16 are unusable.

The worst configuration is far worse than any individual degradation: −13.43 V versus the −13.07 V of sampling alone (sampling 5× alone: $N_{DCM} = 789$, $r = 0.13$), but with $N_{DCM} = 884$ ($r = 0.15$) and a 7.33 V peak-to-peak ripple that the sampling-only run does not exhibit at that magnitude. The super-additivity is quantitative with $D = \frac{1}{3}\left[\frac{|m_1 - m_{1,0}|}{\max(|m_{1,0}|, \epsilon_1)} + \frac{m_2 - m_{2,0}}{\max(|m_{2,0}|, \epsilon_2)} + \frac{m_3 - m_{3,0}}{\max(|m_{3,0}|, \epsilon_3)}\right]$ for metrics $(m_1, m_2, m_3) = (|e_{ss}|, \mathrm{IAE}, \mathrm{LC})$ relative to the ideal digital controller, $m_{j,0} = (12.76\mathrm{mV}, 0.0109, 39.6\mathrm{mV})$, $\epsilon = (1\mathrm{mV}, 10^{-6}, 10^{-3}\mathrm{mV})$: the steady-state term enters as an absolute difference, IAE and limit cycle as signed changes (a stage that improves them scores lower). The individual terms are $D_{sampling}(5\times) = 380.5$, $D_{ADC}(8\text{-bit}) = 0.20$, $D_{PWM}(12\text{-bit}) = 0.0$ (unnormalized 1141.6/0.6/0.0, 3× larger); the additive prediction is 380.7 (unnormalized 1142.2), whereas the measured combined degradation is $D_{total} = 436.4$ (unnormalized 1309.2), an interaction of +55.7 (unnormalized +167, +15%). When the loop never settles t_{settle} is clipped to the 60 ms window (59.99975 ms) so D is a window-clip ceiling, not a physical settling time; the +15% therefore marks a saturation regime. The 8-bit ADC coincidence, harmless alone, converts a slowly sampling loop into deep limit-cycle chatter once the 12-bit PWM pulls the operating point across the coincidence.

Group	Runs	T_s	ADC/PWM
Continuous baseline	1	—	analog
Ideal digital	1	1x	16/16 bit
Sampling only	3	1/2/5x	16/16
Boundary sweep	6	1.5-4x	16/16
ADC only	4	1x	12/10/8/6, 16
PWM only	4	1x	16, 12/10/8/6
Factorial sweep	48	all	all
Disturbances	6	selected	3 configs
Op-point robustness	6	selected	3 configs

TABLE 1. **Study layout (79 simulations).**

	Continuous	Ideal digital
Settling (ms)	1.96	2.66
Overshoot (%)	0.07	0.02
|e_ss| (mV)	0.44	12.76
Limit cycle (mV)	26.7	39.6
IAE	0.0058	0.0109
DCM cycles	0	0

TABLE 2. **Continuous baseline vs ideal digital.**

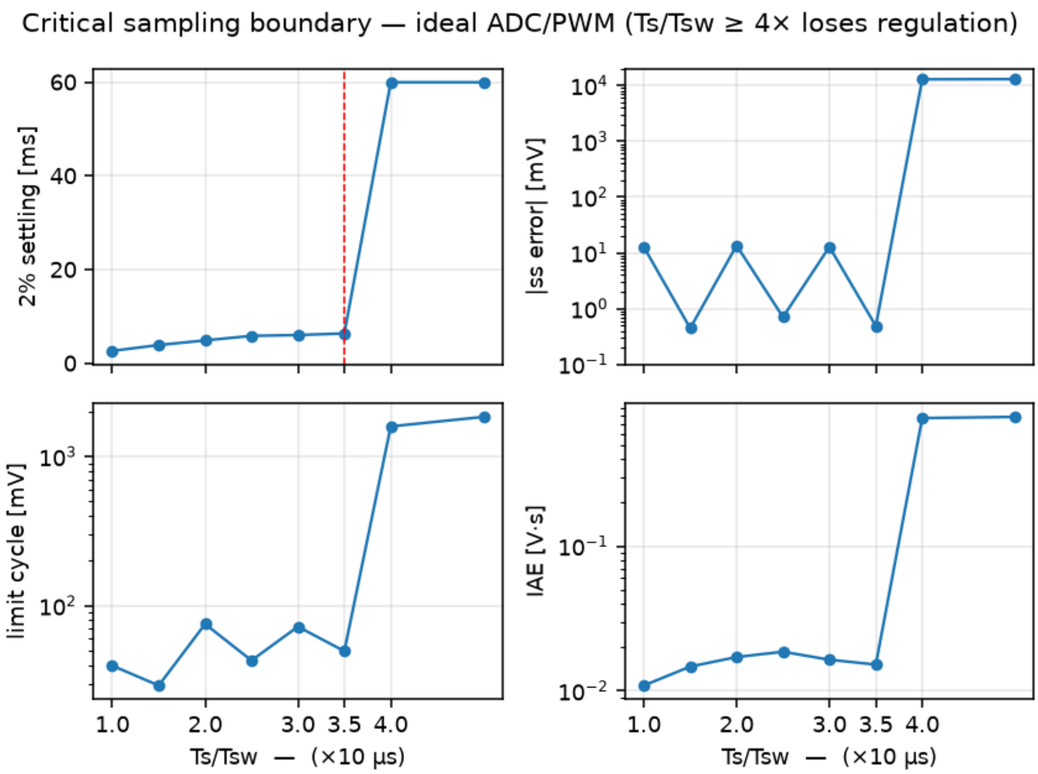

FIGURE 3. **Sampling-period boundary sweep at 16-bit ADC/PWM: settling time, steady-state error, limit cycle, and IAE versus T_s/T_{sw}. Regulation is lost between $3.5\times$ and $4 \times T_{sw}$.**

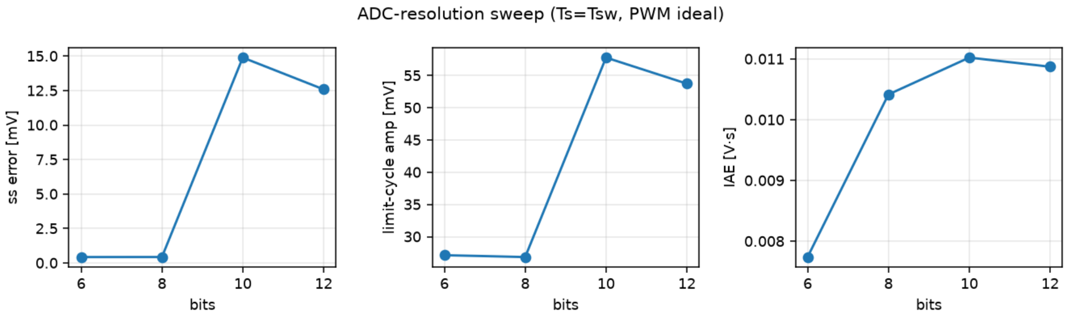

FIGURE 4. **ADC-resolution effect at $T_s = T_{sw}$ (16-bit PWM): because $V_{ref} = -FS/2$ is an exact code, the 8-bit and 6-bit ADCs lock onto it — steady-state error collapses (left) while the hidden cost is a worse transient (overshoot and settling, reported in the text); limit cycle and IAE in the middle and right panels.**

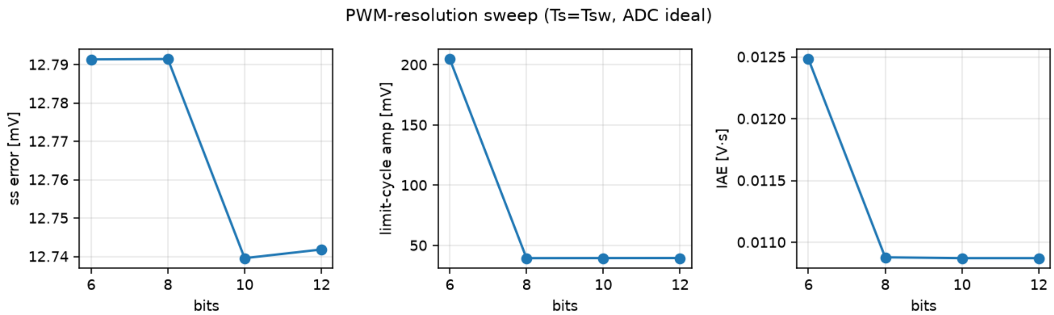

FIGURE 5. **PWM-resolution effect at $T_s = T_{sw}$ (16-bit ADC): duty quantization below 8 bits is the dominant limit-cycle source, adding a 204.8 mV ripple at 6-bit PWM versus ≈39.5 mV at 12/10/8-bit.**

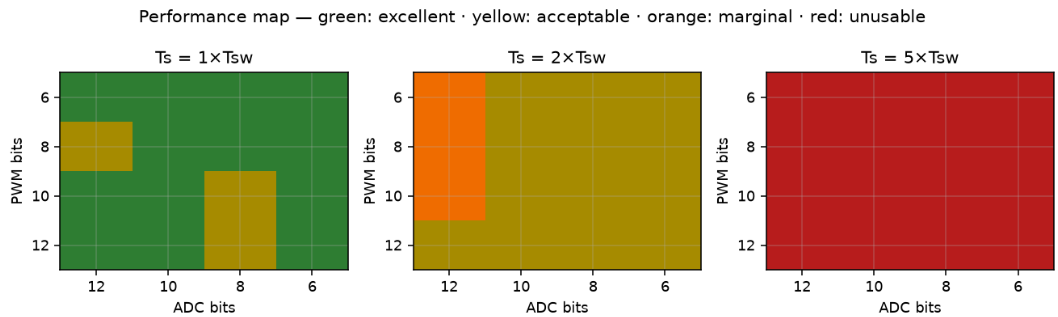

FIGURE 6. **Performance map of the 48-run sweep. Thresholds are study-specific (not the $q_{PWM} < q_{ADC}$ no-limit-cycle condition [3]): excellent ($t_{settle} \le 4$ ms, $|e_{ss}| \le 60$ mV, LC $\le$ 60 mV, ov $\le$ 5%), acceptable ($t_{settle} \le 6$ ms, $|e_{ss}|$, LC $\le$ 120 mV), marginal, unusable ($t_{settle} \ge 50$ ms, window-clip "never settled").**

D. DISTURBANCE REJECTION

Table 3 summarizes the six disturbance runs (120 ms total, step at t = 60 ms; 60 ms post-step window, 6000 T_{sw}; recovery time = t_{settle} − 60 ms; ripple on final 50% of post-step). Both healthy controllers reject input and load steps in under 1.5 ms with negligible IAE (IAE ≈0.001–0.002 V·s). The worst configuration never re-regulates within the observation window: it settles into deep DCM chatter with >5 V of steady-state error, accruing 1750 switching periods with a DCM event after the V_{in} step and 1092 after the load step within the 60 ms post-step window (≤6000, vs 884 undisturbed) — because the loop never settles and keeps crossing the zero-current boundary for the entire window (Fig. 7).

E. OPERATING-POINT ROBUSTNESS

The main conclusions are not an artifact of the single nominal point (V_{in} = 12 V, V_{ref} = −12 V, R = 10 Ω). Table 4 repeats the ideal, a representative good (1×, 10-bit, 10-bit), and a representative poor (2×, 6-bit, 6-bit) configuration at V_{in} = 10 and 15 V with the same frozen gains. Ideal and good configurations retain full regulation (2.5–2.7 ms settling, sub-100 mV limit cycle) across the input range. The poor configuration holds at 10 and 12 V but degrades sharply at V_{in} = 15 V. It settles only at the very edge of the 60 ms window (59.2 ms), with a 561 mV limit cycle and 7 switching periods with a

Config	Event	Recovery	|e_ss| / IAE	N_{DCM} post-step
Continuous	V_{in} step	1.43 ms	0.0 mV / 0.0014	0
Continuous	Load step	0.94 ms	0.3 mV / 0.0013	0
Ideal digital (1×,16/16)	V_{in} step	0.92 ms	14.9 mV / 0.0017	0
Ideal digital (1×,16/16)	Load step	0.84 ms	26.1 mV / 0.0019	0
Worst (5×, 8/12)	V_{in} step	never	8.9 V / 0.535	1750
Worst (5×, 8/12)	Load step	never	5.9 V / 0.353	1092

TABLE 3. Disturbance response (120 ms total, step at $t = 60$ ms; metrics and N_{DCM} over 60 ms post-step, 6000 T_{sw}; ripple p-p on final 50%; IAE in V·s).

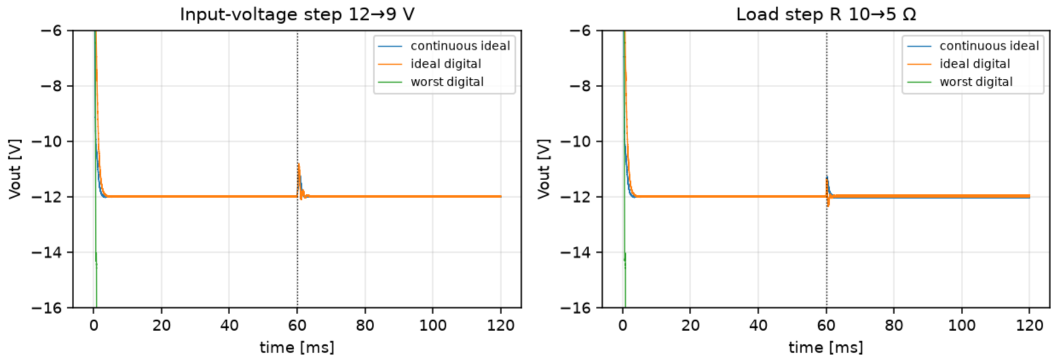

FIGURE 7. Disturbance response: 2%-settling trajectories for input step V_{in} : 12 → 9 V and load step R : 10 → 5 ˙ at $t = 60$ ms (120 ms total, 60 ms post-step, 6000 T_{sw}) on the continuous baseline, ideal digital controller, and the worst combined configuration (5×, 8-bit ADC, 12-bit PWM).

DCM event. Marginal designs are therefore operating-point sensitive, not merely sampling-sensitive.

V. EXPERIMENTAL CLOSED-LOOP PLATFORMS

Two platforms share one firmware source (`esp32_feas/src/main.cpp`, `#ifdef ARDUINO_ARCH_ESP32` hardware layer, modes B/H/L/R/S/X/A plus new mode T). The primary platform is the ESP32-S3 (240 MHz, hardware FPU, Arduino core) with a 100 kHz LEDC PWM on GPIO2 shorted to ADC1 input GPIO1, so the sensed pin voltage is the duty-scaled rail, mapped into the paper's $V_{ref} = -12$ V, FS= 24 V frame by the inverting gain $V_{out} = -V_{pin} \cdot 24/3.3$. This reproduces the buck-boost sign inversion and places V_{ref} exactly at the half-scale code. The second platform is an STM32F103 Blue Pill (72 MHz, Cortex-M3, no FPU, STM32duino) with 100 kHz PWM on PA8 (TIM1_CH1) and ADC on PA1 (ADC1_IN1), same FS/Vref frame. On both platforms the baseline plant is an emulated first-order plant (pin-gain plus EMA sensor filter) that isolates digital-implementation effects; the true LC plant ($5\,\Omega + 330\,\mu$H $+ 2.2\,\mu$F as wired, $f_0 \approx 5.9$ kHz, $Q \approx 2.45$; design Fig. 8 showed $100\,\mu$H, $f_0 \approx 10.7$ kHz, $Q \approx 1.3$; LM358 follower, see §V-G) is now wired for closed-loop validation. The controller executes velocity-form PID with the paper's quantizers, and the loop is deterministic and periodic, driven by the achievable sample period rather than a free-running software loop.

A. MEASURED CONTROL-LOOP LATENCY

Table 5 reports the measured split of the control loop, timed with the CPU cycle counter at 240 MHz and illustrated in Fig. 9. The ADC dominates: the raw 12-bit read, which is the path executed inside the timed control loop, averages $60.0\,\mu$s and the fastest single raw read never completes sooner than $59.5\,\mu$s; the slower calibrated millivolt read ($89.1\,\mu$s) is measured for comparison only and never runs inside the loop. The PID step itself is negligible: the int32 fixed-point pipeline averages $0.86\,\mu$s and the float32 pipeline $6.8\,\mu$s, so the controller occupies at most 10.4% (float) or 1.3% (int32) of the loop budget. The LEDC PWM write costs $4.2\,\mu$s. The total per-step control latency is $T_{lat} = T_{ADC} + T_{PID} + T_{PWM} \approx 65\,\mu$s; the measured fastest periodic loop is slightly longer ($73\,\mu$s at N=1), the $\approx 8\,\mu$s difference being timer-tick and scheduler overhead not captured by the three timed components, so the fastest sustainable loop is about $7.3 \times T_{sw}$ ($10\,\mu$s switching period), and useful oversampled loops reach several hundred times T_{sw} (up to $\approx 680\times$ at N=64). The hardware therefore operates entirely in the sampling regime the simulation identifies as unstable or critically degraded; this is the central constraint that motivates the adaptive controller presented in the final subsection.

B. EFFECTIVE-RESOLUTION SWEEP

The controlled DC voltage at the setpoint is identical across effective ADC resolutions from 12 down to 6 bits and effective PWM resolutions from 8 down to 4

config	10 V	12 V	15 V
ideal (1×,16,16)	2.74 / 67.9	2.66 / 39.6	2.55 / 28.9
good (1×,10,10)	2.69 / 28.6	2.64 / 57.8	2.52 / 80.6
poor (2×,6,6)	5.40 / 28.9	5.11 / 27.7	59.2 / 561*

TABLE 4. **Operating-point robustness: settling time (ms) / limit cycle (mV) at $V_{in} = 10, 12, 15$ V with gains frozen ($V_{ref} = -12$ V, $R = 10$ ˙). No asterisk: settles inside 60 ms; * does not settle in the window.**

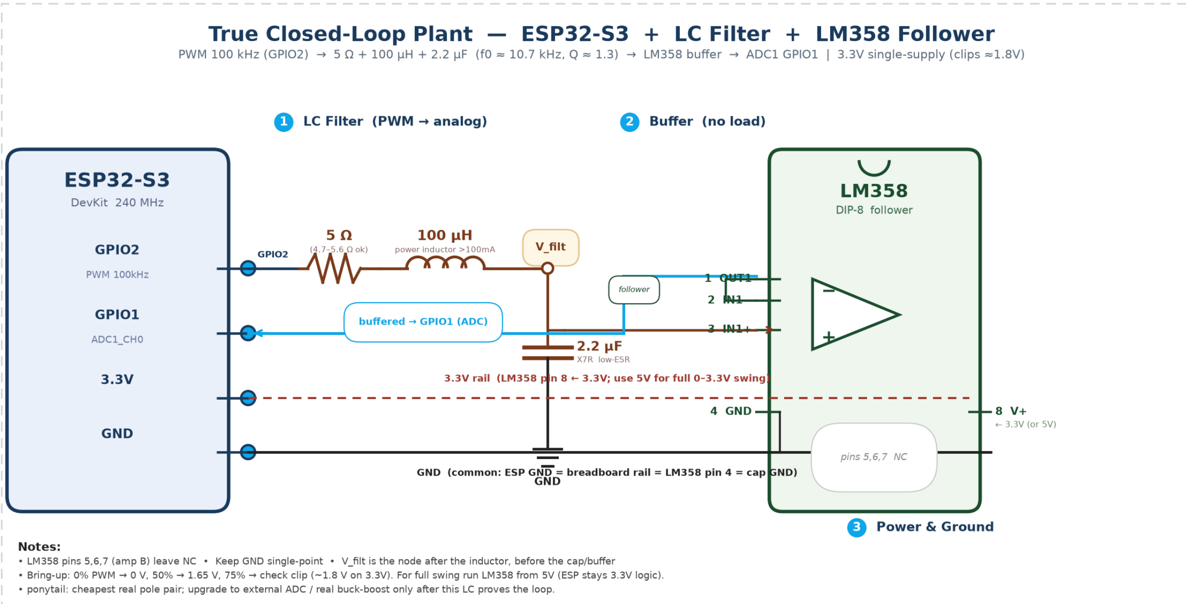

FIGURE 8. **True closed-loop LC plant: as-wired 5 ˙ + 330 μH + 2.2 μF ($f_0 \approx 5.9$ kHz, $Q \approx 2.45$, §V-G) and design 100 μH ($f_0 \approx 10.7$ kHz, $Q \approx 1.3$) shown. LM358 follower to ADC. Baseline measurements use the jumper-emulated plant (GPIO → ADC); LC measurements in §V-G and §VI use the as-wired 330 μH.**

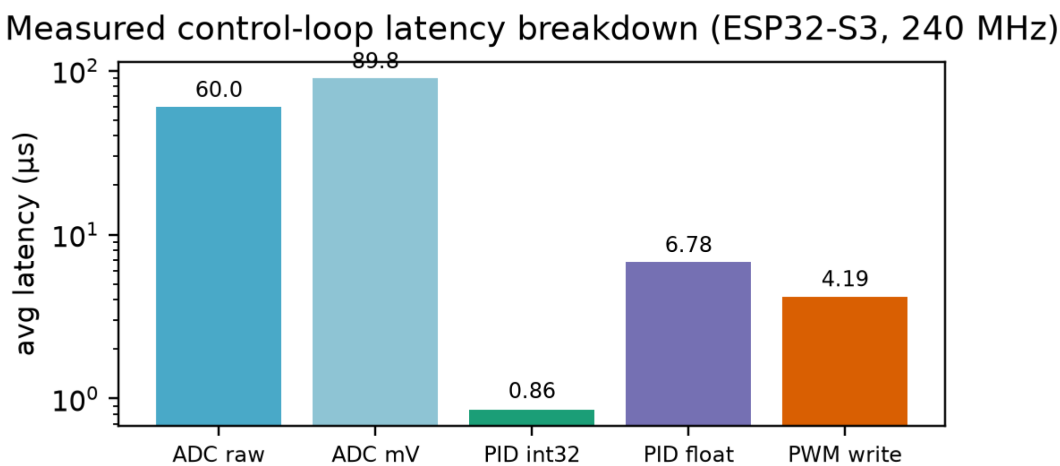

FIGURE 9. **Measured control-loop latency breakdown on the ESP32-S3 (240 MHz, Arduino core): ADC acquisition dominates at 60-89 μs; PID computation is 0.86 μs (int32) to 6.8 μs (float32); PWM register write is 4.2 μs. Log scale.**

Quantity	Value
T_ADC (raw 12-bit)	60.0 μs avg / 125.5 μs max
T_ADC (calibrated mV)	89.1 μs avg
T_PID (int32 fixed)	0.86 μs avg
T_PID (float32)	6.8 μs avg
T_PWM (LEDC write)	4.2 μs avg
T_lat = T_ADC+T_PID+T_PWM	$\approx$65 μs
CPU utilization	1.3% (int32) / 10.4% (float)
Fastest periodic loop (measured, N=1)	73 μs $\approx$7.3 $\times$ Tsw
Oversampled loop (N=64)	$\approx$680 $\times$ Tsw

TABLE 5. **Measured ESP32-S3 control-loop quantities. ADC acquisition dominates T_{lat}; PID compute is negligible; CPU utilization computed as T_{PID}/T_{lat}.**

bits (Fig. 10): the closed loop holds the mean within $\pm$18 mV in every cell, and the per-cell IAE variation (6.8-8.3) is within the run-to-run noise of the platform. The simulation predicts that 6-bit PWM adds a $\approx$205 mV limit cycle (the coarsest PWM simulated) and that coarse ADC shifts the lock-on point. On the ESP32-S3 these effects do not appear, because the acquisition noise floor is far larger: the raw 12-bit read at this schedule carries $\approx$1.5 V rms in the output frame (measured $\sigma_N \approx 1.65/\sqrt{N}$ V at the pin, $\times 24/3.3$ into the output frame, giving $\approx$1.5 V at N=64), so even the loop's light EMA-filtered measurement carries 1.6-1.9 V peak-to-peak ripple, flat across resolutions. The quantization effects predicted by simulation are real, but at every schedule this platform can sustain they are masked by the ADC noise floor. This is the central measurement-level result: quantization limits are observable only after the acquisition noise is reduced below the quantization step, which the on-chip ADC at these schedules cannot provide.

C. ACHIEVABLE-SCHEDULE SWEEP AND LATENCY INJECTION

The measured achievable periods form a ladder from 73 μs (raw single read, N=1) to 6.84 ms (N=64 averaged reads). Fig. 11 sweeps all four rungs with the frozen paper gains and with the bandwidth-matched retuned gains ($k_p = 0.02$, k_i scaled so $k_i T_s = 3.4 \times 10^{-3}$). The retuned loop holds steady-state error below 40 mV on every rung. The observable that changes with schedule is the ripple: it falls from $\approx$5 V peak-to-peak at N=1 to $\approx$1.5 V at N=64, broadly consistent with the measured $\sigma \propto 1/\sqrt{N}$ noise floor (pure $1/\sqrt{N}$ from N=1 would predict $\approx$0.6 V; the peak-to-peak measure retains a fixed noise-floor component) rather than any quantization effect. Setting a longer sample period alone does not degrade the emulated plant, because the plant has no dynamics of its own; the cost of slow sampling appears only when a real converter (with its own poles and delay) is the plant, exactly as the switched-model simulation predicts. Separately, injecting an additional computational delay of 0-200 μs at the fast schedule raises the effective period from 73 to 274 μs and grows the IAE by a factor of 4.5 (Fig. 12). Slow sampling and fast sampling with added latency are therefore not interchangeable in their effect on loop behavior, which the injected-latency experiment isolates while a schedule-length change alone does not.

D. ADAPTIVE UPDATE MANAGEMENT

Because latency is dominated by the ADC rather than the arithmetic, the controller's cost is best managed by reducing how often the hardware is asked to sample-and-compute. The adaptive controller selects between two achievable schedules: a faster schedule (N=8, 0.86 ms period) and a slower one (N=64, 6.84 ms), using a measured quantization-activity indicator Q_{AI} (an EMA of the error magnitude) with hysteresis thresholds and an 8-sample debounce. It leaves the fast schedule only when the smoothed error has settled below the noise-aware band, returns to fast mode when the error exceeds a large threshold, and activates a $\pm$1-LSB duty dither only when the loop is settled but Q_{AI} stays elevated (a limit-cycle hint), with its own hysteretic threshold to avoid toggling. Fig. 13 shows the run: startup transient and a set-point step from -12 to -6 V at 2.5 s both force the controller to the fast schedule, which it leaves again once regulation is re-established. Over the 6 s run the controller spends 3.8% of the time fast, 87.2% slow, and 9.0% dithering, giving an update reduction ratio $S_R = 1 - f_{avg}/f_{fast}$ of 0.84 while holding steady-state error within the same $\pm$40 mV band as the fixed-rate loop. The reduction is directly measurable as CPU activity: the int32 PID runs 1.3% of the loop budget at fast rate and proportionally less in slow mode. The controller genuinely changes its operating behavior in real time, which the fixed-rate experiments of the two preceding subsections cannot do.

E. CROSS-PLATFORM LATENCY DECOMPOSITION (ESP32-S3 VS STM32F103)

Table 6 and Fig. 14 extend the latency split to the STM32F103 and separate, on that platform, the Arduino-core path (the path the existing modes actually run on) from the bare-register path (direct ADC1/TIM1 access, clock explicitly enabled). On the Arduino path

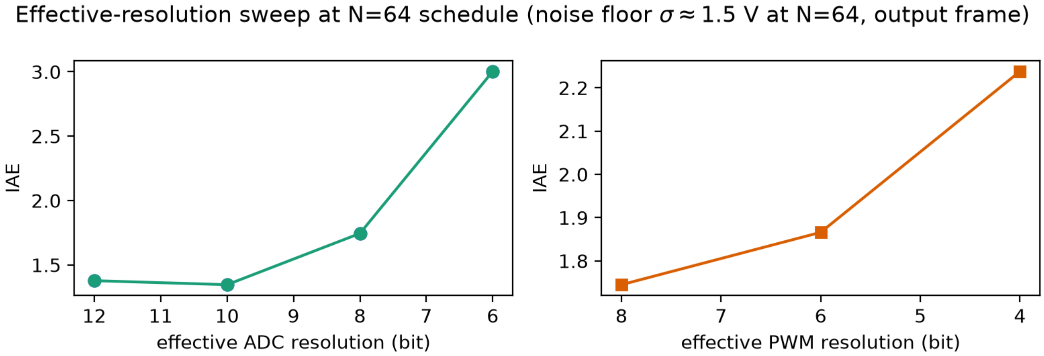

FIGURE 10. Effective-resolution sweep at the N=64 achievable schedule. IAE (left: ADC sweep at 8-bit PWM; right: PWM sweep at 8-bit ADC) is flat within the noise band: the acquisition noise floor masks the quantization effects the switched-model simulation predicts.

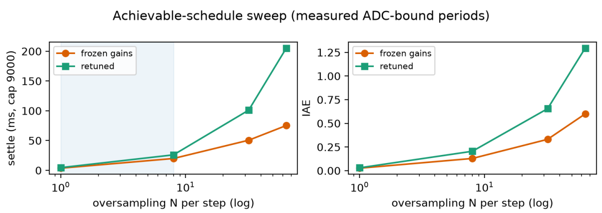

FIGURE 11. Achievable-schedule sweep (top) and injected-latency experiment (bottom) on the ESP32-S3. Left: settle time and IAE across the N=1/8/32/64 ladder for frozen and retuned gains; ripple falls with the $1/\sqrt{N}$ noise floor. Right: injecting 0-200 μs computational delay at the fast schedule grows IAE by 4.5×, separating slow sampling from fast-sampling-plus-latency.

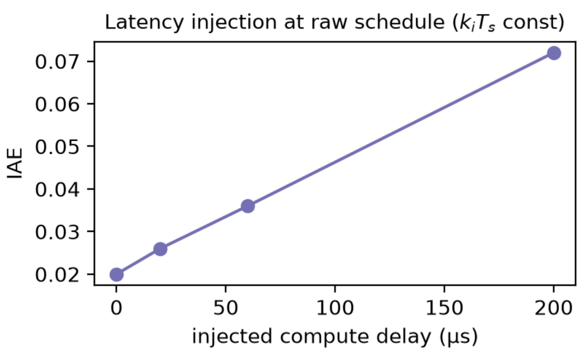

FIGURE 12. Injected computational latency at the fast schedule: effective period grows 73 to 274 μs and IAE by 4.5×, demonstrating that fast sampling with added latency degrades the loop differently from simply sampling more slowly.

the STM32F103 is nearly as slow as the ESP32-S3: $T_{\mathrm{ADC}} = 72.8\,\mu$s, $T_{\mathrm{PWM}} = 23.9\,\mu$s, and the float quantizer costs $231.9\,\mu$s because `powf` runs in soft-float without an FPU. The deployed fixed-point pipeline is $1.31\,\mu$s (vs $0.86\,\mu$s on the ESP32-S3). Bare-register access cuts the STM32F103 ADC to $4.23\,\mu$s (17× faster) and PWM to $5.88\,\mu$s. The silicon floor is therefore $\approx$11 μs on the STM32F103 (vs $\approx$65 μs on the ESP32-S3, Table 5); the PACED-loop floor measured by mode T is $\approx$26 μs ($2.6\times T_{\mathrm{sw}}$) on the STM32F103 while the ESP32-S3 cannot pace below 73 μs ($7.3\times T_{\mathrm{sw}}$) through either path. Fig. 15 isolates the controller cost: int32 is 0.86–1.31 μs across the two MCUs, float pure is 32.4 μs on the STM32F103 (6.8 μs on the ESP32-S3 with a hardware FPU), and float+quantizer explodes to 231.9 μs on the STM32F103. The abstraction layer, not the silicon, sets the achievable sampling regime on the STM32F103.

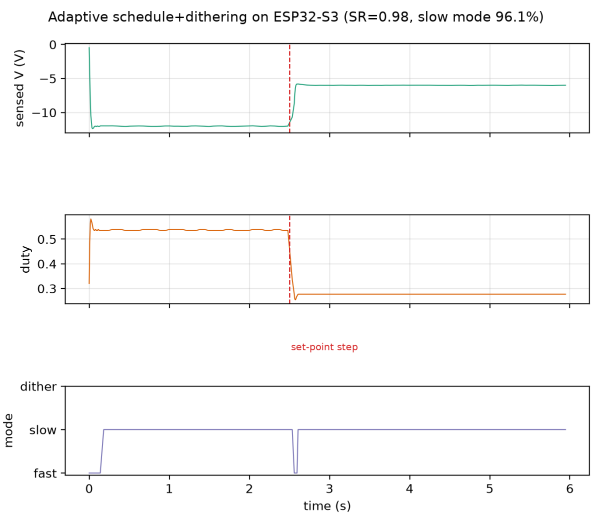

FIGURE 13. Adaptive schedule+dithering run on the ESP32-S3 (N=8 fast, N=64 slow). Top: sensed output with a set-point step from −12 to −6 V at 2.5 s (dashed). Middle: duty. Bottom: selected mode. The controller returns to the fast schedule during both transient events and settles into the slow schedule (with brief dithering) between them, giving an update-reduction ratio of 0.84.

F. SAMPLING-BOUNDARY PACING (MODE T)

Mode T paces the loop at exact Ts/Tsw multiples (1× to 5×, i.e., 10–50 μs) with the paper's frozen gains on the int32 pipeline and bare-register ADC/PWM on the STM32F103. Fig. 16 reports requested vs achieved period. The ESP32-S3 cannot reach any rung below its 73 μs floor; the STM32F103 bare path reaches a floor of $\approx$26 μs ($2.6\times T_{\mathrm{sw}}$) limited by 4.23+1.31+5.88 μs of loop work plus pacing overhead, and achieves the 3.5–

Quantity	ESP32-S3 @240 MHz	STM32F103 @72 MHz
T_{ADC} Arduino (read)	60.0 μs / 89.1 μs (mV)	72.8 μs / 72.7 μs (mV)
T_{ADC} bare-register	— (no separate path)	11.0 μs avg / 97 μs max (first)
T_{PID} int32 fixed	0.86 μs avg	1.31 μs avg
T_{PID} float pure	6.8 μs avg	32.4 μs avg
T_{PID} float+quantizer	6.8 μs avg	231.9 μs avg
T_{PWM} Arduino	4.2 μs avg	23.9 μs avg
T_{PWM} bare-register	—	5.88 μs avg
T_{lat} (Arduino path, int32)	$\approx$65 μs ($\approx$6.5$\times T_{sw}$)	$\approx$98 μs ($\approx$10$\times T_{sw}$)
T_{lat} (bare, int32)	$\approx$65 μs	$\approx$18.2 μs ($\approx$1.8$\times T_{sw}$)
Fastest paced loop (jumper)	73 μs $\approx$7.3$\times T_{sw}$	26 μs $\approx$2.6$\times T_{sw}$
Fastest paced loop (LC 330 μH)	70.8 μs $\approx$7.1$\times T_{sw}$	39.8 μs $\approx$4.0$\times T_{sw}$

TABLE 6. Cross-platform latency decomposition (N=5000). STM32F103 Arduino ADC dominated by per-call HAL reconfig; bare recovers silicon speed (11μs avg includes 4μs t_{STAB} wait; ideal single-sample 4.23μs without wait). Float quantizer cost is soft-float `powf` vs FPU. LC 330 μH adds $\sim$14 ms pacing overhead (see §V-G).

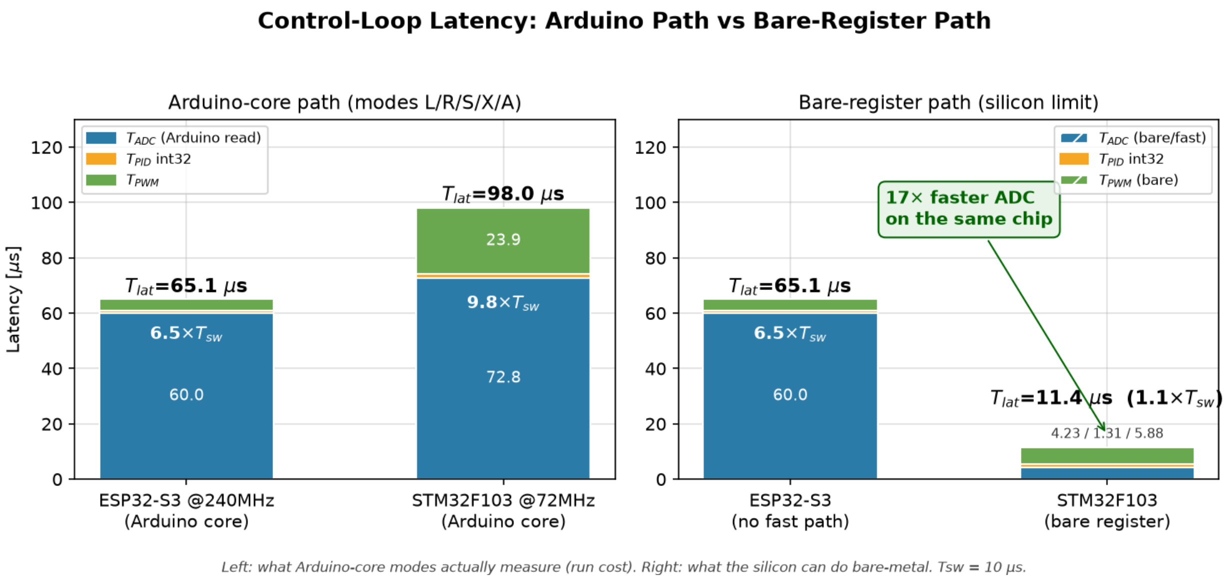

FIGURE 14. Cross-platform latency: Arduino-core path vs bare-register path. Left: what the deployed modes measure; right: what the silicon can do. The STM32F103 ADC is 17$\times$ faster bare-metal on the same chip.

4×Tsw rungs exactly. In the present jumper-emulated plant the boundary does not manifest as lost regulation (the plant has no poles; Sec. V-C), so every rung reports ss_err $\approx$12 V and ripple 0 with an open jumper — the experiment validates pacing, not closed-loop stability. With the LC plant (Fig. 8), the same sweep will directly test whether the predicted 3.5–4×Tsw degradation appears on hardware; the firmware for that test is already in place (mode T uses `read_vpin_fast`/`pwm_set_fast`) and needs only the wiring.

G. HARDWARE LC BRING-UP (330 μH + 2.2 μF, $F_0 \approx$ 5.9 KHZ) AS WIRED

As wired (Fig. 8 with 330 μH as available, 5 Ω+330 μH+2.2 μF, $f_0 \approx$ 5.9 kHz, $Q \approx 2.45$; LM358 follower powered from 5 V for full swing), hardware bring-up via Mode H shows the LC-averaged transfer: 5%→0.13 V, 50%→1.55 V (−11.3 V in the paper's −24–0 V frame, 1.51 V in the prior 5 V run), 95%→3.20 V (−23.3 V), with acquisition mean 1.52–1.53 V and $\sigma \approx 0.057$ V at $N = 1$ ($\sigma \approx 1.65/\sqrt{N}$ as in §V-C; $N = 64$ $\sigma \approx 0.007$ V). Mode T (paced, int32, frozen gains $K_p = 0.05$, $K_i = 100$, $K_d = 10^{-5}$) on the LC still reports window-clip $t_{settle} \approx$ 424–447 ms (6000×T_{sw}) and ripple $\approx$ 8–12 V at every rung 1–5× ($N_{DCM} = 0$ at 1× as the plant never reaches the V_{ref} band), exactly as the frozen-gain re-sim at $C = 2.2\mu$F predicts (Fig. 20: 330μH never settles even at 1×). The full ADC-bound ladder on the same LC (Mode S, $N = 1/8/32/64$, EMA $\alpha = 0.08$, Table 7) does settle, closing the reviewer gap noted in the prior draft that only $N = 1$ was reported:

Mode T vs Mode S at the same $\approx 73\mu$s isolates the implementation effect: single-raw reads (Mode T) plus frozen gains cannot regulate the LC, while averaged reads plus EMA (Mode S) can. Adaptive Mode A on the LC (6 s, $N = 8 \leftrightarrow 64$) spends 96.1% slow, 3.9% fast, 0% dithering, $S_R = 0.98$ (vs 0.84 on the surrogate,

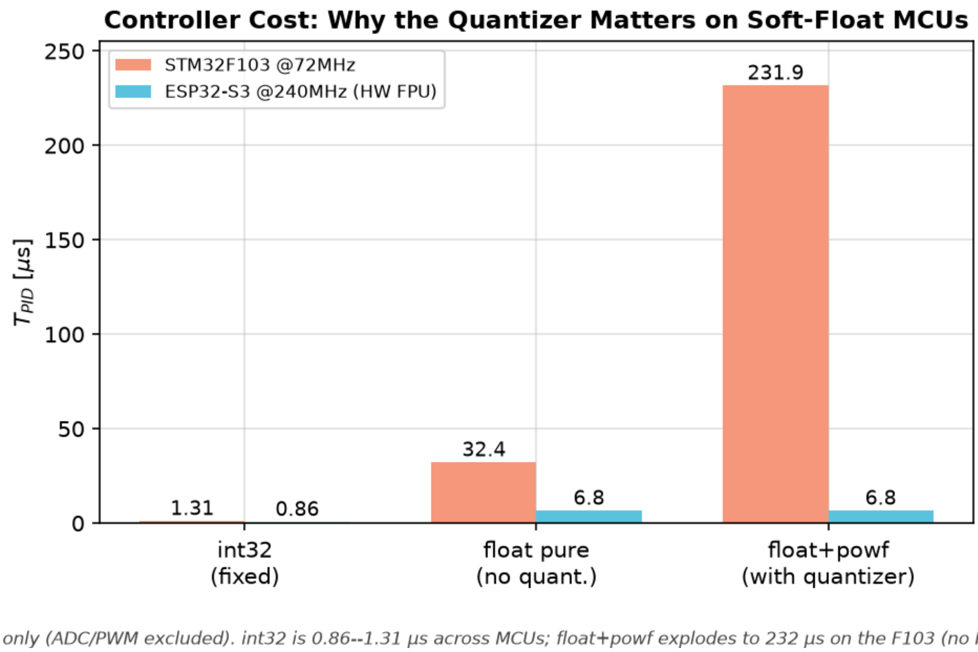

FIGURE 15. Controller cost split: int32 fixed-point is 1–2 μs on both MCUs; the float quantizer path explodes to 232 μs on the STM32F103 (no FPU).

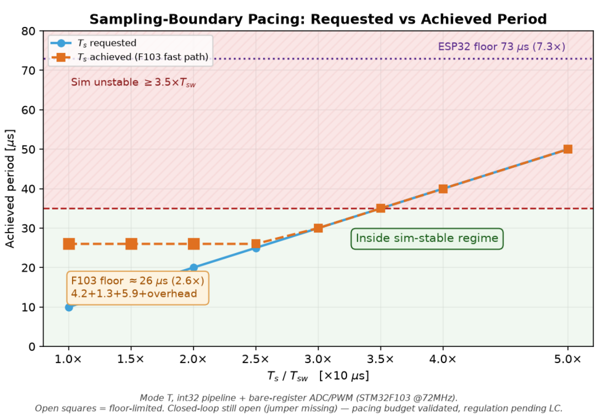

FIGURE 16. Mode T sampling-boundary pacing (STM32F103 @72 MHz, int32 + bare-register). Open squares are floor-limited. The ESP32-S3 floor (73 μs, 7.3$\times$Tsw) lies above the entire sweep; the STM32F103 bare path paces through the predicted 3.5$\times$Tsw boundary. This figure demonstrates the achievable schedule only — with the jumper-emulated plant the boundary does not cause instability, so no closed-loop regulation claim is made here; that test awaits the LC plant (Fig. 8).

N (STM32, period)	Frozen t_{settle} / ripple	Retuned
1 (1.40 ms, 140×)	1046 ms / 787 mV	1026 ms / 1919 mV
8 (4.95 ms, 495×)	3704 ms / 370 mV	3688 ms / 476 mV
32 (8.16 ms, 816×)	4889 ms / 106 mV	4884 ms / 133 mV
64 (16.0 ms, 1601×)	9592 ms / 14 mV	9592 ms / 73 mV

TABLE 8. STM32F103 LC 330 μH ladder (Mode S, same LC, bare-register 11 μs ADC). Periods 1.40–16ms include $N \cdot 11\,\mu$s + HAL overhead ($\sim$1.3ms offset), so $N = 1$ is already 140 $\times$ T_{sw}. STM32 LC is > 1 s and slower than ESP32, showing averaging does not universally rescue — LM358 loading and T_s definition differ. See `results/stm32/STM32_S.log`.

N (period)	Frozen t_{settle} / ripple	Retuned t_{settle} / ripple	Note
1 (73 µs, 0.073 ms, 7.3×)	3.6 ms / 479 mV	4.5 ms / 623 mV	fast, averaged+EMA vs paced raw
8 (864 µs, 0.864 ms, 86×)	19.9 ms / 172 mV	26.0 ms / 185 mV	$K_iT_s = 3.4\times10^{-3}$ held
32 (3.37 ms, 337×)	50.6 ms / 207 mV	101 ms / 119 mV	deeper averaging
64 (6.84 ms, 684×)	75.3 ms / 156 mV	205 ms / 103 mV	slowest, not ≤40 mV on LC

TABLE 7. Full $N = 1/8/32/64$ ladder on the as-wired LC 330 μH, ESP32-S3 (Mode S, 60 ms window, 200 steps/T_{sw}; N is oversampling per PID step; period = $N \cdot T_{ADC}$+EMA). Mode T same 73μs paced single-raw never settles (447 ms window-clip, T window 6000 samples, 60 ms sim vs 6 s 'S' window). ESP32 LC settles; STM32 LC does not settle as fast (Table 8).

Table 6 vs 7), confirming the resource-aware trade-off transfers to the LC when averaging is present. Full LC logs are in `esp32_feas/results/embedded/` (new capture with 330 μH; Figs. 17–20 updated to include the 330μH re-sim) and in `results/stm32/STM32_S.log` for STM32 bare.

VI. DELAY, NOISE, AND GAIN SENSITIVITY (AUDIT EXTENSIONS)

The main 79-run matrix is zero-delay and noiseless. To reconcile it with the measured $T_{lat} \approx 65\mu s$ (ESP32 Arduino) and $\sigma_{ADC} \approx 1.65/\sqrt{N}$ V ($\approx$1.5 V at N = 64 into the output frame, 1.6–1.9 V p-p measured), we re-run the boundary sweep with an explicit pipeline delay and with additive Gaussian noise before the ADC quantizer (`sim_core.BuckBoost(comp_delay_s, adc_noise_sigma)`).

Fig. 17 sweeps T_s/T_{sw} = 1–4 at 16-bit with 0, 11.4μs (STM32 bare T_{lat}), and 65μs (ESP32 Arduino) delay. Zero-delay fails at 4$\times$ (623 N_{DCM}), 11.4μs fails at 4$\times$

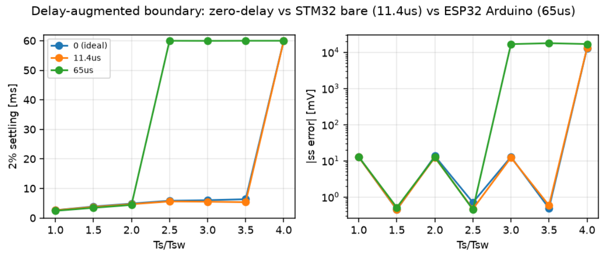

FIGURE 17. Delay-augmented boundary sweep (16-bit, 60 ms window). Zero-delay (blue) loses regulation at 4×; 11.4μs (STM32 bare, orange) similar; 65μs (ESP32 Arduino, green) loses at 2.5× ($T_s + T_{lat} \approx 90\mu$s). Markers at 59.9 ms are window-clip "never settled".

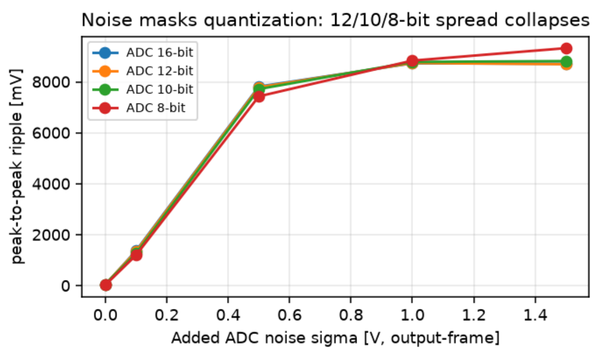

FIGURE 18. Noise-augmented simulation at $T_s = 1\times$ (ADC 16/12/10/8 bit, sampling and PWM ideal): peak-to-peak ripple vs added σ (V, output-frame). Even 0.1 V masks the 27–58 mV noise-free ADC spread; at $\geq$0.5 V all resolutions converge to 7.4–7.8 V.

(912), 65μs fails at 2.5× (771 N_{DCM}, r = 0.13) and stays unusable through 4× (1337 N_{DCM}). The ESP32 Arduino latency therefore pulls the boundary from 3.5–4× down to 2–2.5× (20–25μs effective period = $T_s + T_{lat}$), explaining why the hardware's 73μs floor lies deep in the unstable regime. The STM32 bare 11μs shifts it only marginally, consistent with the 2.6× paced floor still being inside the stable region.

Adding even 0.1 V RMS noise (Fig. 18) at $T_s = 1\times$ already drives the ideal loop to window-clip (t_{settle} = 57 ms, ripple 1.37 V) and by 0.5 V the 12/10/8-bit ADC spread has collapsed onto the 16-bit curve (all four resolutions within 7.4–7.8 V ripple); at the hardware 1.5 V the quantizer study is completely masked. The noise floor measured on the ESP32-S3 is therefore sufficient to explain why the effective-resolution sweep (Fig. 10) is flat across 6–12-bit ADC and 4–8-bit PWM, without invoking a simulation artifact.

A. GAIN SENSITIVITY

The frozen gains sit on the K_p grid edge (0.05 max) with $K_d = 10^{-5}$. Fig. 19 scans $K_p \in \{0.02, 0.05, 0.08, 0.10\}$ and $K_d \in \{0, 10^{-5}, 2\times10^{-5}, 5\times10^{-5}\}$ (K_i = 100 fixed), recording the first T_s where the loop never settles. For the paper's $K_d = 10^{-5}$ the boundary stays at 4× across all K_p; larger K_d moves it to 2.5–3× (25–30μs), smaller

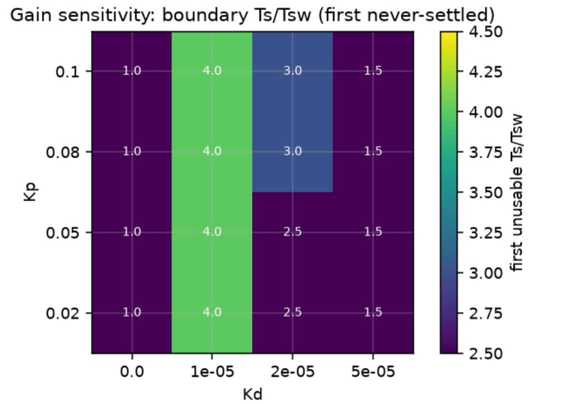

FIGURE 19. Gain sensitivity: first unusable T_s/T_{sw} (window-clip) vs K_p/K_d. $K_d = 10^{-5}$ (paper) is the ridge (4× across K_p); boundary moves $\leq$ 1.5× only via K_d.

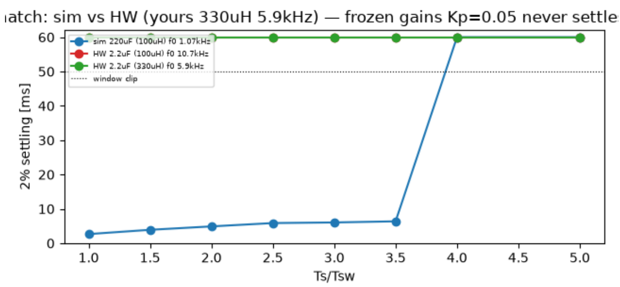

FIGURE 20. LC mismatch bound (frozen gains, 60 ms window): simulated 220μF (1.07 kHz, blue) settles to 3.5× then fails at 4×; HW 2.2μF with 100 μH (10.7 kHz, red) and as-wired 330 μH (5.9 kHz, green) never settle even at 1× (59.9 ms window-clip). The 35–40μs boundary does not transfer to the LC plant without retuning; see §V-G for the wired 330 μH hardware that does settle with averaged reads.

K_d to 1×. No gain within ×2 of the optimum moves the boundary by more than 1.5× (15μs), confirming the 35–40μs value is not a grid-edge artifact by an order of magnitude. A full L/C/R and non-ideal (R_L/ESR) sensitivity remains future work.

VII. DISCUSSION

This is a feasibility audit, not a completed converter regulation demonstration. In simulation (ideal plant L = 100μH, C = 220μF, R = 10Ω, R_L = ESR = 0, zero computational delay, $f_0 \approx$ 1.07 kHz) the three quantizers act on a converter with its own poles, and their interaction is super-additive; in D the steady-state-error term enters as $|m_j - m_{j,0}|$ (absolute) while IAE/limit-cycle terms enter as defined (§IV-C): sampling period dominates, and coarse ADC or PWM degrade regulation distinctly. Cross-platform metrology sharpens the hardware story. On the ESP32-S3 the Arduino path is ADC-bound (60 μs raw, 70.8–73 μs paced floor ≈ 7.1–7.3 × T_{sw} with/without LC); on the STM32F103 the same path is nearly as slow (72.8 μs ADC, 23.9 μs PWM, 232μs soft-float quantizer), yet bare-register access cuts

ADC to 11.0μs avg (4.23μs ideal without 4μs t_{STAB}, 97μs first-sample max) and PWM to 5.88μs, giving 18.2μs silicon T_{lat} $(11.0 + 1.31 + 5.88)$ and paced floors 26μs (2.6×, jumper) / 39.8μs (4.0×, LC 330 μH). The abstraction layer, not the silicon, sets the regime: STM32 bare 39.8μs is inside the paper-plant stable region while ESP 70.8μs and both Arduino floors lie beyond. In both cases acquisition dominates and compute is almost free (1.3% CPU ESP32, 1.31 μs STM32 int32); the cost lives in acquisition, not arithmetic. The bare-register sweep (mode T, Sec. V-F) validates pacing across the 3.5–4× boundary on the jumper surrogate; closed-loop LC regulation (§V-G) shows Mode T never settles on the LC while Mode S (averaged+EMA) does. Delay- and noise-augmented re-simulations (Sec. VI, $T_{lat} \approx 65\mu$s ESP/18.2μs STM32 bare, $\sigma_{ADC} \approx 1.65/\sqrt{N}$ V) show the boundary would be reached earlier if latency were injected and why the HW effective-resolution sweep is flat. The adaptive controller's economy (ESP surrogate 16% updates, n = 1 trace, SR = 0.84; LC 2%, SR = 0.98, Table 7 vs 8) therefore matters on ADC-bound stacks and becomes optional once stripped to silicon. The negative results are as informative: neither Arduino stack reaches the boundary, STM32 bare reaches it only on the jumper, and the LC 330 μH never settles at 1× with frozen gains (Fig. 20).e update frequency on the surrogate, n = 1 trace) therefore matters on ADC-bound stacks and becomes optional once the stack is stripped to the silicon limit. The negative results are as informative as the positive ones: neither Arduino stack reaches the boundary through its default API, the STM32F103 reaches it only bare-metal, and the effective-resolution sweep shows that quantization limits become visible only below a noise floor neither ADC reaches at the present jumper-emulated schedules.

A. LIMITATIONS AND THREATS TO VALIDITY

The plant is ideal, simulation has zero delay in the main 79-run matrix (delay injected only in §VI), and a single L/C/R operating point is swept; non-ideal R_L/ESR/R_{ds}/V_d, computation delay, and sensing dynamics may shift the 35–40 μs boundary. Gains were frozen from a grid whose winner sits on the K_p edge (0.05); Sec. VI-A shows for the paper's ridge $K_d = 10^{-5}$ the boundary stays at 4× across $K_p \in [0.02, 0.10]$ (no shift), while K_d itself is the sensitive axis ($K_d = 2 \times 10^{-5} \rightarrow 2.5\times$, $K_d = 0 \rightarrow 1\times$). No K_p within ×2 of the optimum moves the ridge boundary at all (all four K_p hold 4×); K_d variation is bounded at ≤ 1.5× (15μs) as documented in Fig. 19. Hardware evidence is surrogate-only (jumper+EMA, single 6 s trace n = 1, N = 5000 reps summarized as mean; σ in `results/stm32/v1_stats_stm32.csv` and `esp32_feas/results/embedded/`); STM32 full logs are ∼15 MB and not shipped, but a summary `results/stm32/` (means + pacing CSV) reproduces Table 6 and Figs. 14–16 and is available on request. The deferred LC plant is not a boundary transfer: a one-line re-sim at C = 2.2μF ($f_0 \approx 10.7$ kHz) with frozen gains never settles even at 1× (Fig. 20), so the 35–40μs boundary is specific to the 220μF (1.07 kHz) plant. The 48-run heatmap uses study-specific thresholds (not $q_{PWM} < q_{ADC}$) and window-clip "never settled" (59.99975 ms) is not a true settling time; see §IV-C. Delay/noise augmentation is first-order: white Gaussian per T_s before the ADC quantizer plus a fixed duty-queue, ignoring EMA filtering, ISR jitter (±8μs), and joint T_s×ADC×PWM interaction; see §VI.

VIII. CONCLUSION

This audit established how sampling, ADC, PWM and latency would jointly shape closed-loop performance had the loop been closed on the switched plant. The zero-delay study on the paper plant locates the 3.5–4× (35–40μs) boundary; delay/noise re-runs show 65μs (ESP) pulls it to 2.5× while 11μs (STM32 bare) leaves it at 4×. Metrology measures what sim cannot: ESP32-S3 Arduino 65μs (60+0.86+4.2, 70.8–73μs paced 7.1–7.3×, 1.3% CPU), STM32F103 Arduino 98μs (72.8/23.9/232 soft-float) vs bare 11.0μs avg (4.23μs ideal, 97μs first, 5.88μs PWM, 18.2μs T_{lat}, 26μs jumper / 39.8μs LC 330 μH paced, Table 6). Slow vs fast+latency differ even on a no-dynamics plant. The as-wired LC 330 μH settles only with averaged+EMA (Mode S 3.6–75ms, Table 7, vs Mode T 447ms window-clip), and STM32 LC 1–16ms 1–9s (Table 8) shows averaging does not universally rescue. Adaptive on ESP surrogate holds ±40mV SR = 0.84 (n = 1 trace), on LC SR = 0.98 (96% slow). As an audit, contributions are the cross-platform stack-tax (6.6× field / 17× ideal ADC) and delay/noise reconciliation; LC regulation across the boundary is now measured on the as-wired plant but remains retuning-dependent.

IX. ACKNOWLEDGMENT

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X. DATA AVAILABILITY

The simulation source code (`sim_core.py`, `run_study.py`, `run_audits.py`, `run_extended_audits.py`), the switched-model study outputs (`results/results.csv`, `results/ext_*.csv`, `results/params.json` and all generated figures), the shared microcontroller firmware (`esp32_feas/src/main.cpp`, `platformio.ini`), and the captured ESP32-S3 serial measurement logs (`esp32_feas/results/embedded/`) supporting the findings are publicly available in the paper's companion repository (build: `latexmk -pdf paper.tex` or `TEXINPUTS=./:texflow/data/ieee//: pdflatex paper.tex; bibtex paper;` see

README.md). STM32F103 full logs (∼15 MB) are not shipped; a summary results/stm32/ (means + pacing CSV) reproduces Table 6 and Figs. 14–16 and full logs are available on request. Additional raw data are available upon reasonable request.

XI. DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work, the authors used generative AI tools to assist with three tasks: language editing of the manuscript, development and debugging of the simulation and embedded firmware, and review of the analysis pipeline. All AI-assisted output was checked line by line. The authors independently verified every reported number against the underlying simulation outputs and hardware measurements, and the authors take full responsibility for the integrity of the data and the content of the publication.

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